THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

School of Professional Education and Executive Development
專業進修學院

SPEED

SEHS4654 Capstone Project

Student Handbook

Bachelor of Engineering (Honours) in Electrical Engineering

Programme Code: 84066 (Full-time)/ 84065 (Part-time)

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1. Introduction

This document aims at providing you with the necessary information on the formal requirements of the SEHS4654 Capstone Project and setting out guidelines on how the Capstone Project reports should be prepared, completed and presented.

The Capstone Project is a significant part of your Honours assessment, which accounts for 6 credits in the programme of Bachelor of Engineering (Honours) in Electrical Engineering.

1.1. Objectives

To provide students with an opportunity of integrating and applying knowledge from different disciplines of electrical engineering to conduct an engineering project that is open-ended and requires team collaboration for its completion.

1.2. Programme Intended Learning Outcomes (PILOs)

(A) Professional/Academic Knowledge and Skills

(P-ILO-A1) apply fundamental principles and knowledge of mathematics, science and engineering to the areas of electrical engineering and related disciplines.

(P-ILO-A2) design and conduct experiments with appropriate techniques and tools, and interpret and analyse the data.

(P-ILO-A3) design a system, component or process according to specifications and requirements, and identify the realistic constraints, such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, which may influence engineering systems, projects or solutions.

(P-ILO-A4) identify, formulate and reason critically alternative views or solutions to solve practical problems in the areas of electrical engineering and related disciplines.

(P-ILO-A5) keep abreast of the latest developments in electrical engineering and related disciplines.

(P-ILO-A6) understand professional and ethical responsibility.

(P-ILO-A7) understand different cultures and social development needs and acquire knowledge of contemporary issues and the broad education necessary to realise the impact of engineering solutions in the local, national, global and societal contexts, especially the importance of health, safety and environmental considerations to both workers and the general public.

(P-ILO-A8) use the techniques, skills, and modern engineering tools necessary for engineering practice.

(P-ILO-A9) use the computational/IT tools along with an understanding of their processes and limitations.

(B) Attributes for All-roundedness

(P-ILO-B1) communicate effectively via graphic, numeric, verbal and written media with proficiency in both English and Chinese.

(P-ILO-B2) show awareness on the need for and to engage in lifelong learning.

(P-ILO-B3) work in multi-disciplinary teams with professional interpersonal skills.

1.3. Subject Intended Learning Outcomes (SILOs)

Upon successful completion of this subject, you will be able:

(a) apply specialised knowledge independently and as a team;
(b) communicate progress and achievement orally and in written reports;
(c) formulate a creative, rich in intellectual content and challenging project to solve key engineering problems;
(d) examine the progress of a project from concept to final implementation and testing, through problem definition and the selection of alternative solutions;
(e) synthesise relevant knowledge and analytical skills gained in various engineering domains;
(f) develop professionalism with self-confidence and independence by successfully completing the project in a competent manner.

1.4. How Learning Outcomes are accomplished

Specific assessment methods/ tasks	**% weighting**	**Intended subject learning outcomes to be assessed**					
		a	**b**	**c**	**d**	**e**	**f**
Continuous Assessment	**100%**						
1. Assessment of student's performance	10%	✓			✓		✓
2. Formal project proposal	5%		✓	✓			
3. Interim progress report	20%		✓	✓	✓		
4. Final report	40%	✓	✓	✓	✓	✓	✓
5. Presentation and demonstration	25%	✓	✓				✓
Total	100%						

1. Performance of each student should be assessed individually together with the team's overall performance by the supervisor, an independent assessor, the peers and a project assessment panel consisting of several academic staff (both full-time and part-time programmes usually use the same panel).
2. The student's performance in a project group as a whole, and that of each group member on an individual basis, are assessed by the supervisor. The formal project proposal is assessed by the supervisor. The interim progress report and the final report are assessed by both the supervisor and the independent assessor. As part of the assessment process, each group member is required to specify his/her own contribution in completing the project when compared to his/her teammates (peer assessment).
3. The supervisor monitors and assesses the overall and individual progresses through regular meetings. The formal project proposal is submitted at around week 4 of the first semester. The interim report is submitted at around week 19 of the first semester. The final report is submitted before the end-of-year examination. The presentation and demonstration will be conducted before the end-of-year examination also. Individual contribution and performance of each student in performing the project will be assessed.
4. During the presentation and demonstration, every group member is required to present the project especially on his/her significant contribution to the whole project and respond to the questions addressed to him/her by the examination panel. Both group and individual performance will be assessed.
5. The assessment system is summarized as shown in the following table:

Assessor	**Assessment Component (% of the total)**				
	Student's Performance (10%)	**Formal Project Proposal (5%)**	**Interim progress report (20%)**	**Final report (40%)**	**Presentation and demonstration (25%)**
Supervisor	✓	✓	✓	✓	
Independent Assessor			✓	✓	
Examination Panel					✓

2 Group Formation

The Capstone Project is a group project. Each group should generally be consisted of 3 members. In exceptional cases where formation of 3-member groups are practically impossible, 2-member or 1-member groups are allowed with special approval from the programme leader. Group formation should be done by the students and it will be coordinated by the project coordinator. Each group should appoint a Project Manager who will provide overall leadership for the project. Other members should also play key leadership roles during project execution phase.

3 Project Proposals

A set of project proposals is made available for students to select for their capstone project. The proposals are generally drawn from the academic staff of CPCE and the industry. Each project is supervised by at least one project supervisor. The proposals are generally fall into 2 categories:

a. Research-based proposals: Involve experiments, design and development of test rigs, computer simulations, data analysis and interpretations
b. Design-based proposals: Involve design of products/systems, computer simulations and analysis, prototype building and testing, data analysis and interpretations

Students are also encouraged to propose their own project proposals. Such proposals should be written with project title, objectives, brief description and estimated budget. Project coordinator will screen such proposals in consultation with the programme leader and other academic staff. If suitable, students are allowed to carry out the project and a suitable project supervisor is assigned for supervision.

When selecting the given proposals, students are advised to consider preferences of all the group members, skills of individual members and also the group size. If necessary, the students may contact the respective project supervisors for further details of the projects. Each group should select 10 proposals and indicate them in the online project selection form in the order of preference. Submission deadlines will be announced by the project coordinator.

Project allocations for student groups is done by considering 3 factors as follows:

i. Average GPA of all team members
ii. Order of preferences provided by the student groups
iii. Number of projects allocated for each academic supervisor

Efforts will be made to allocate most preferred projects to student groups as far as practically possible.

Students are required to submit a formal project proposal with literature review, and scheme design or research plan.

4 Role of the Supervisor

Each project will have at least one supervisor. This subject aims at developing your ability in independent problem solving and critical thinking. The supervisor's role is to:

- Provide appropriate guidance;
- Comment on the progress and quality of your work;
- Give advice to the best of his/ her ability;
- Arrange the first meeting with industrial advisor/s;
- Provide assistance in arranging lab facilities (if required); and
- Perform assessment on your Continuous Progress and Final Project Report.

It is NOT your Supervisor's responsibility to:

- Proofread your written reports or any drafts;
- Perform any pre-marking of your written reports;
- Specify what you need to do or tell you what to do; and
- Manage your work schedule or progress

5 Role of Students

To carryout capstone project independently and to make the best use of supervisor's support, you should:

- Manage your work progress and schedule properly;
- Initiate communication and arrange regular meetings with your supervisor/s;
- Prepare a clear agenda for each meeting with the supervisor/s;
- Prepare and keep clear records of the discussion (in the form of a log sheet);
- Honour the explicit and implied confidentiality agreements with the industrial advisor/s
- As a good practice, maintain a logbook that records your design ideas, meeting records with your groupmates/supervisors/lab staff, raw data from your lab experiments, etc.
- Ask for comments but NOT direct instructions from your supervisor.

The final responsibility of carrying out your group's Capstone project and developing intended learning outcomes resides with you but NOT your supervisor.

6 Assessment Criteria and Rubrics

6.1 Grading

Subject Grade	*Grade Point*	*Short Description*	*Elaboration on subject grading description*
A+	*4.3*	*Excellent*	***Demonstrates excellent achievement of intended subject learning outcomes by being able to skillfully use concepts and solve complex problems. Shows evidence of innovative and critical thinking in unfamiliar situations, and is able to express the synthesis or application of ideas in a logical and comprehensive manner.***
A	*4.0*		
A-	*3.7*		
B+	*3.3*	*Good*	***Demonstrates good achievement of intended subject learning outcomes by being able to use appropriate concepts and solve problems. Shows the ability to analyse issues critically and make well-grounded judgements in familiar or standard situations, and is able to express the synthesis or application of ideas in a logical and comprehensive manner.***
B	*3.0*		
B-	*2.7*		
C+	*2.3*	*Satisfactory*	***Demonstrates satisfactory achievement of intended subject learning outcomes by being able to solve relatively simple problems. Shows some capacity for analysis and making judgements in a variety of familiar and standard situations, and is able to express the synthesis or application of ideas in a manner that is generally logical but fragmented.***
C	*2.0*		
C-	*1.7*		
D+	*1.3*	*Pass*	***Demonstrates marginal achievement of intended subject learning outcomes by being able to solve relatively simple problems. Can make basic comparisons, connections and judgements and express the ideas learnt in the subject, though there are frequent breakdowns in logic and clarity.***
D	*1.0*		
F	*0.0*	*Fail*	***Demonstrates inadequate achievement of intended subject learning outcomes through a lack of knowledge and/or understanding of the subject matter. Evidence of analysis is often irrelevant or incomplete.***

Indicative descriptors for modifier grades:

Main Grade (solid)	The student generally performed at this level, indicating mastery of the subject intended learning outcomes at this level.
+ (exemplary)	The student consistently performed at this level and exceeded the expectations of this level in some regards, but not enough to claim mastery at the next level.
- (marginal)	The student basically performed at this level, but the performance was inconsistent or fell slightly short in some regards.

Note: The above indicative descriptors for modifier grades are not applicable to the passing grades D and D+

'F' is a subject failure grade, whilst all others ('D' to 'A+') are subject passing grades. No credit will be earned if a subject is failed.

6.2 Assessment Rubrics

Formal Project Proposal (5%)

Subject Intended Learning Outcomes (SILOs) to be achieved:

b. communicate progress and achievement orally and in written reports.

c. formulate a creative, rich in intellectual content and challenging project to solve key engineering problems.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Introduction: *Background, problem identification, objectives (10%)*	Rationale for carrying out the project is well explained. Aims and objectives are explicitly stated.	Rationale for carrying out the project is explained. Aims and objectives are quite well stated.	Rationale for carrying out the project is mostly explained, but need some improvements. Aims and objectives are stated.	Rationale for carrying out the project is briefly explained and need considerable improvements. Aims and objectives are stated briefly.	Rationale for carrying out the project is incomplete/missing. Aims and objectives are incomplete/missing.
Literature Review / Market Research: *Relevance and comprehensiveness (20%)*	Literature review/market research is comprehensive, relevant, consistent with the project topic and critically analysed.	Literature review/market research is mostly comprehensive, relevant, consistent with the project topic and generally analysed.	Literature review/market research is partly relevant, sometimes inconsistent with the project topic and briefly analysed.	Literature review/market research is partly relevant, mostly inconsistent with the project topic and briefly described.	Literature review/market research is mostly irrelevant, inconsistent with the project topic and very briefly description.
Technical content: *Background theory, research/design methodology, design content (design of concept solutions, test rigs, experiments ...), technical skills and analysis (40%)*	Appropriate research/design methods are used. Detailed & challenging engineering analysis at all the stages. Excellent technical skills are demonstrated.	Mostly appropriate research/design methods are used. Detailed & challenging engineering analysis at most of the stages. Good technical skills are demonstrated.	Somewhat appropriate research/design methods are used. Limited engineering analysis at some stages. Satisfactory technical skills are demonstrated.	Research/design methods are partially relevant/appropriate. Very Limited engineering analysis. Very limited technical skills are demonstrated.	Research/design methods are irrelevant/inappropriate. No engineering analysis. No/Very limited technical skills are demonstrated.
Project Management and Progress: *Project schedule, resource planning, leadership (10%)*	Work scope is justifiable and workable. Provide sufficient evidence of excellent resource management and leadership. Show an excellent progress.	Work scope is mostly justifiable and workable. Provide sufficient evidence of good resource management and leadership. Show a good progress.	Work scope is mostly justifiable and workable with some modifications. Provide some evidence of resource management and leadership. Show a satisfactory progress.	Work scope is partly justifiable and workable with considerable modifications. Provide limited evidence of resource management and leadership. Show some progress.	Work scope is unjustifiable and not workable. Provide no evidence of resource management and leadership. Show very limited progress.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Overall Presentation: *Format, quality of technical writing, references (20%)*	Report is very well-organized, comprehensive & consistent with the requirements. Almost free from grammatical & spelling errors. References are clear and correct.	Report is quite well-organized, comprehensive & consistent with the requirements. Few grammatical & spelling errors. Most references are clear and correct.	Report is somewhat organized & mostly consistent with the requirements. Some grammatical & spelling errors. Some references are clear & correct.	Report is not well-organized & sometimes inconsistent with the requirements. Quite a lot of grammatical & spelling errors. Only few references are clear & correct.	Report is not organized & inconsistent. Lots of grammatical & spelling errors. References are missing/unclear & incorrect.

Interim Progress Report (20%)

Subject Intended Learning Outcomes (SILOs) to be achieved:

b. communicate progress and achievement orally and in written reports.
c. formulate a creative, rich in intellectual content and challenging project to solve key engineering problems.
d. examine the progress of a project from concept to final implementation and testing, through problem definition and the selection of alternative solutions.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Introduction: *Background, problem identification, objectives (10%)*	Rationale for carrying out the project is well explained. Aims and objectives are explicitly stated.	Rationale for carrying out the project is explained. Aims and objectives are quite well stated.	Rationale for carrying out the project is mostly explained, but need some improvements. Aims and objectives are stated.	Rationale for carrying out the project is briefly explained and need considerable improvements. Aims and objectives are stated briefly.	Rationale for carrying out the project is incomplete/missing. Aims and objectives are incomplete/missing.
Literature Review / Market Research: *Relevance and comprehensiveness (20%)*	Literature review/market research is comprehensive, relevant, consistent with the project topic and critically analysed.	Literature review/market research is mostly comprehensive, relevant, consistent with the project topic and generally analysed.	Literature review/market research is partly relevant, sometimes inconsistent with the project topic and briefly analysed.	Literature review/market research is partly relevant, mostly inconsistent with the project topic and briefly described.	Literature review/market research is mostly irrelevant, inconsistent with the project topic and very briefly description.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Technical content: *Background theory, research/design methodology, design content (design of concept solutions, test rigs, experiments ...), technical skills and analysis (40%)*	Appropriate research/design methods are used. Detailed & challenging engineering analysis at all the stages. Excellent technical skills are demonstrated.	Mostly appropriate research/design methods are used. Detailed & challenging engineering analysis at most of the stages. Good technical skills are demonstrated.	Somewhat appropriate research/design methods are used. Limited engineering analysis at some stages. Satisfactory technical skills are demonstrated.	Research/design methods are partially relevant/appropriate. Very Limited engineering analysis. Very limited technical skills are demonstrated.	Research/design methods are irrelevant/inappropriate. No engineering analysis. No/Very limited technical skills are demonstrated.
Project Management and Progress: *Project schedule, resource planning, leadership (10%)*	Work scope is justifiable and workable. Provide sufficient evidence of excellent resource management and leadership. Show an excellent progress.	Work scope is mostly justifiable and workable. Provide sufficient evidence of good resource management and leadership. Show a good progress.	Work scope is mostly justifiable and workable with some modifications. Provide some evidence of resource management and leadership. Show a satisfactory progress.	Work scope is partly justifiable and workable with considerable modifications. Provide limited evidence of resource management and leadership. Show some progress.	Work scope is unjustifiable and not workable. Provide no evidence of resource management and leadership. Show very limited progress.
Overall Presentation: *Format, quality of technical writing, references (20%)*	Report is very well-organized, comprehensive & consistent with the requirements. Almost free from grammatical & spelling errors. References are clear and correct.	Report is quite well-organized, comprehensive & consistent with the requirements. Few grammatical & spelling errors. Most references are clear and correct.	Report is somewhat organized & mostly consistent with the requirements. Some grammatical & spelling errors. Some references are clear & correct.	Report is not well-organized & sometimes inconsistent with the requirements. Quite a lot of grammatical & spelling errors. Only few references are clear & correct.	Report is not organized & inconsistent. Lots of grammatical & spelling errors. References are missing/unclear & incorrect.

<u>Final Report (40%)</u>

Subject Intended Learning Outcomes (SILOs) to be achieved:

a. apply specialised knowledge independently and as a team.
b. communicate progress and achievement orally and in written reports.
c. formulate a creative, rich in intellectual content and challenging project to solve key engineering problems.
d. examine the progress of a project from concept to final implementation and testing, through problem definition and the selection of alternative solutions.
e. synthesise and apply their knowledge and analytical skills gained in various engineering domains.
f. develop professionalism with self-confidence and independence by successfully completing the project in a competent manner.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Introduction: *Background, problem identification, objectives (5%)*	Rationale for carrying out the project is well explained. Aims and objectives are explicitly stated.	Rationale for carrying out the project is explained. Aims and objectives are quite well stated.	Rationale for carrying out the project is mostly explained, but need some improvements. Aims and objectives are stated.	Rationale for carrying out the project is briefly explained and need considerable improvements. Aims and objectives are stated briefly.	Rationale for carrying out the project is incomplete/missing. Aims and objectives are incomplete/missing.
Literature Review / Market Research: *Relevance and comprehensiveness (10%)*	Literature review/market research is comprehensive, relevant, consistent with the project topic and critically analysed.	Literature review/market research is mostly comprehensive, relevant, consistent with the project topic and generally analysed.	Literature review/market research is partly relevant, sometimes inconsistent with the project topic and briefly analysed.	Literature review/market research is partly relevant, mostly inconsistent with the project topic and briefly described.	Literature review/market research is mostly irrelevant, inconsistent with the project topic and very briefly description.
Technical content: *Background theory, research/design methodology, design content (design of concept solutions, test rigs, experiments …), technical skills and analysis (40%)*	Appropriate research/design methods are used. Detailed & challenging engineering analysis at all the stages. Excellent technical skills are demonstrated.	Mostly appropriate research/design methods are used. Detailed & challenging engineering analysis at most of the stages. Good technical skills are demonstrated.	Somewhat appropriate research/design methods are used. Limited engineering analysis at some stages. Satisfactory technical skills are demonstrated.	Research/design methods are partially relevant/appropriate. Very Limited engineering analysis. Very limited technical skills are demonstrated.	Research/design methods are irrelevant/inappropriate. No engineering analysis. No/Very limited technical skills are demonstrated.
Results/output and Discussion: *Data handling, interpretations & evaluations (for research projects), details of the final output and testing (for design projects) (20%)*	Results are effectively interpreted and critically discussed with consistent focus on the aim. Complete set of details for product/system realization is provided with detailed tests & evaluations.	Results are interpreted and discussed with consistent focus on the aim. Details for product/system realization is provided with appropriate tests & evaluations.	Results are satisfactorily interpreted and discussed with somewhat consistent focus on the aim. Some details for product/system realization are provided with some tests & evaluations.	Partial information on results and interpretation of data. Some details for product/system realization are provided with no/limited tests & evaluations.	Results and interpretation are missing or incomplete. Insufficient details for product/system realization are provided with no tests & evaluations.
Conclusion and Recommendations: *Achievement of aims & objectives, realistic recommendations (5%)*	Conclusion addresses the research question/issue or design need and achievement of aims & objectives. Detailed and realistic recommendations for future development.	Conclusion mostly addresses the research question/issue or design need and achievement of aims & objectives. Good recommendations for future development.	Conclusion somewhat addresses the research question/issue or design need and achievement of aims & objectives. Satisfactory recommendations for future development.	Conclusion barely addresses the research question/issue or design need and achievement of aims & objectives. Barely satisfactory recommendations for future development.	Conclusion is incomplete and does addresses the research question/issue or design need and achievement of aims & objectives. No/unrealistic recommendations for future development.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Overall Presentation: *Format, quality of technical writing, references (20%)*	Report is very well-organized, comprehensive & consistent with the requirements. Almost free from grammatical & spelling errors. References are clear and correct.	Report is quite well-organized, comprehensive & consistent with the requirements. Few grammatical & spelling errors. Most references are clear and correct.	Report is somewhat organized & mostly consistent with the requirements. Some grammatical & spelling errors. Some references are clear & correct.	Report is not well-organized & sometimes inconsistent with the requirements. Quite a lot of grammatical & spelling errors. Only few references are clear & correct.	Report is not organized & inconsistent. Lots of grammatical & spelling errors. References are missing/unclear & incorrect.

<u>Presentation and Demonstration (25%)</u>

Subject Intended Learning Outcomes (SILOs) to be achieved:

a. apply specialised knowledge independently and as a team.
b. communicate progress and achievement orally and in written reports.
e. synthesise and apply their knowledge and analytical skills gained in various engineering domains.
f. develop professionalism with self-confidence and independence by successfully completing the project in a competent manner.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Organization and Structure (group): *Clarity and logical presentation of information (20%)*	Highly effective introduction brings audience to topic. Well organized content and presented in a logical sequence. A well-designed conclusion matches with the project objectives.	Effective introduction brings audience to topic. Good organized content and presented in a logical sequence. A good conclusion mostly matches with the project objectives.	Satisfactory introduction brings audience to topic sufficiently. Appropriate content and presented mostly in a logical sequence. A satisfactory conclusion mostly matches with the project objectives.	Introduction either underdeveloped or awkward. Somewhat organized, but the content occasionally unclear. Conclusion underdeveloped, or fails to match with the project objectives.	Introduction extremely underdeveloped or missing. Structure is confusing and limited content. Conclusion extremely underdeveloped or missing.
Technical content (group): *Relevance and comprehensiveness (30%)*	Excellent demonstration of full knowledge of the subject with clear explanations and elaborations.	Good demonstration of sufficient knowledge of the subject with appropriate explanations and elaborations.	Satisfactory demonstration of knowledge of the subject with some degree of explanations and elaborations.	Barely Satisfactory knowledge of the subject with limited explanations.	Inadequate knowledge of the subject with very limited explanations.
Design of presentation materials and visual aids (group): *Quality and effectiveness of presentation materials (10%)*	Excellent use of visual aids, very informative, very effective, and not distracting	Good use of visual aids, informative, effective, and not distracting	Satisfactory use of visual aids, mostly informative, mostly effective, and sometimes distracting	Barely satisfactory use of visual aids, barely informative, barely effective, and sometimes distracting	Inadequate use of visual aids, not informative, ineffective, and very distracting

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Presentation skills (individual): *Oral presentation skills, time & pace control (20%)*	Maintains good eye contact, excellent language skills, delivery is poised, controlled, and smooth. Able to complete the presentation right on time	Maintains good eye contact, good language skills, delivery is mostly poised, controlled, and smooth. Able to complete the presentation on time	Maintains eye contact sufficiently, satisfactory language skills, delivery is somewhat poised, controlled, and smooth. Need to rush to complete the presentation on time	Barely maintains eye contact, barely adequate language skills, delivery is poor and not confident. Presentation is too short/too long and unable to use the time allowed effectively.	Very limited eye contact, poor language skills, delivery is poor and not confident at all. Unable to manage the time effectively.
Question & Answers (individual): *Knowledge, quality and relevance of answers (20%)*	Fully prepared, anticipated questions and responded confidently with full information	Mostly prepared, understood questions and responded confidently with sufficient information	Adequately prepared, understood questions but sometimes had difficulty in responding with satisfactory information	Barely prepared, sometimes misunderstood questions, can only answer rudimentary questions	Unprepared, Misunderstood questions and did not respond appropriately

<u>Assessment of Student's Performance (10%)</u>

Subject Intended Learning Outcomes (SILOs) to be achieved:

a. apply specialised knowledge independently and as a team.

d. examine the progress of a project from concept to final implementation and testing, through problem definition and the selection of alternative solutions.

f. develop professionalism with self-confidence and independence by successfully completing the project in a competent manner.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Ability to integrate & apply knowledge from different disciplines and independent learning *(20%)*	Demonstrate excellent ability to integrate and apply knowledge from different disciplines. Acquired new knowledge appropriately and independently.	Demonstrate good ability to integrate and apply knowledge from different disciplines. Acquired new knowledge appropriately and mostly independently.	Demonstrate satisfactory ability to integrate and apply knowledge from different disciplines. Acquired new knowledge satisfactorily with some guidance.	Demonstrate some ability to integrate and apply knowledge from different disciplines. Acquired limited new knowledge with considerable guidance.	Demonstrate no/limited ability to integrate and apply knowledge from different disciplines. Find it difficult to acquire new knowledge even with considerable guidance.

Assessment Criteria	Excellent (Grade: A- to A+)	Good (Grade: B- to B+)	Satisfactory (Grade: C- to C+)	Pass (Grade: D to D+)	Fail (Grade: F)
Use of appropriate engineering tools: *Analytical, experimental and/or computational tools and skills (20%)*	Use variety of analytical, experimental and/or computational tools to support project activities effectively. Showed excellent proficiency in using computational tools.	Use appropriate analytical, experimental and/or computational tools to support project activities. Showed good proficiency in using computational tools.	Use some analytical, experimental and/or computational tools to support project activities. Showed satisfactory proficiency in using computational tools.	Limited use of analytical, experimental and/or computational tools to support project activities. Showed limited proficiency in using computational tools.	No/very limited use of analytical, experimental and/or computational tools to support project activities. Showed no/inadequate proficiency in using computational tools.
Teamwork, professionalism and Communication *(20%)*	Demonstrated professional engagement with supervisor, industrial advisor/s, technical staff and other stakeholders.	Demonstrated good engagement with supervisor, industrial advisor/s, technical staff and other stakeholders.	Demonstrated acceptable engagement with supervisor, industrial advisor/s, technical staff and other stakeholders.	Demonstrated barely acceptable engagement with supervisor, industrial advisor/s, technical staff and other stakeholders.	Unprofessional engagement with supervisor, industrial advisor/s, technical staff and other stakeholders.
Project Management and Continuous Progress: *Project planning, monitoring and control, quality of work (20%)*	Excellent time management. Very detailed action plan. Adequate work accomplished with very good quality.	Good time management. Quite detailed action plan. Adequate work accomplished with reasonable quality.	Acceptable time management. Acceptable action plan. Some work accomplished with reasonable quality.	Barely acceptable time management. Brief action plan. Some work accomplished with acceptable quality.	Fail to manage time & work schedule properly. Incomplete action plan. Very limited work accomplished with barely acceptable quality
Overall Performance: *Attitude, commitment, contribution (20%)*	Excellent working attitude and commitment to achieve set goals. Made significant contributions that far exceeded the expectations in all regards.	Good working attitude and commitment to achieve set goals. Made good contributions that exceeded the expectations in many regards.	Satisfactory working attitude and commitment to achieve set goals. Made satisfactory contributions that meet the expectations in many regards.	Barely satisfactory working attitude and commitment to achieve set goals. Made barely satisfactory contributions that marginally meet the expectations in some regards.	Inadequate working attitude and commitment to achieve set goals. Inadequate contributions that fails to meet the expectations.

7 Guidelines for Formal Project Proposal

The formal project proposal should contain the outputs from preliminary stage of your capstone project, relevant literature reviews and scheme design or research plan.

7.1 Proposal Structure

The format of the proposal will vary depending upon the particular project and this should be discussed with the project supervisor. In general, the proposal should consist of followings and any other extra content as specified by your project supervisor:

- Aims of the project
- Proposed specifications of the product (no matter it is a hardware or software project)
- Summary of the literature search did up-to-date
- Proposed approach/methodology to be used
- Some brief descriptions on the theory of the approach/methodology
- Timetable of your work to the end of the project

The length of the proposal should be limited to about 8-15 pages, excluding appendix.

Title Page: Refer to the sample title page given in the **Appendix A**.

8 Guidelines for Interim Report

The interim report should contain the outputs from initial steps of your capstone project and a detailed plan of activities up to the completion of the project. A major portion of this report may go into your final report. It is advisable to consult the project supervisor and then prepare a plan of the draft report before start writing. The supervisor should be provided with the draft of the interim report well in advance of the submission deadline for his comments and advice.

8.1 Report Structure

The format of the report will vary depending upon the particular project and this should be discussed with the project supervisor. In general, interim report should consist of followings and any other extra content as specified by your project supervisor:

- Title page
- Table of contents
- Introduction
- Aims and objectives
- Methodology
- Literature review/ market research (as appropriate)
- Others (progress made in the project activities organized under appropriate headings)
- Project management (project planning details: Gantt chart, work breakdown and brief evaluation of progress made)
- Conclusion
- References (Use APA referencing style)
- Appendix
    - Log sheets
    - Any other relevant materials as required

8.2 Length of the Report

The report should be written such that it can be read and absorbed by an engineer having a basic knowledge of the subject. It should not contain any irrelevant materials. The report should be clear and concise. It is advisable to write a report with approximately 20 pages of single line spacing (Font: Times New Roman font size 12 pt) at this stage.

Title Page: Refer to the sample title page given in the **Appendix A**.

9 Guidelines for Final Report

The final report should contain the outputs from all the activities undertaken for your capstone project. A considerable portion of this report may come from the interim report. It is advisable to consult the project supervisor and then prepare a plan of the draft report before you start writing. The supervisor should be provided with the draft of the final report well in advance of the submission deadline for his comments and advice.

The format of the report will vary depending upon the particular project and this should be discussed with the project supervisor. The report will be assessed based on elements such as technical content, clarity, layout, standard of academic writing in English, process and analysis of experimental data or simulation results, depth of discussion, useful conclusions, etc.

9.1 Report Structure

The contents of a report shall include:

- Title page
- Capstone Project Declaration Form
- Acknowledgement
- Table of contents
- Abstract
- Nomenclature
- Chapter 1: Introduction
- Chapter 2 – P: The main content of the report
- Chapter Q: Conclusions and possible recommendations for future work
- References
- Appendices
- Log sheets

Title page: A sample of the title page is shown in **Appendix A**.

Capstone Project Declaration Form: A sample of the declaration form is shown in **Appendix B**.

Acknowledgement: This is where you should acknowledge the various parties who initiate, supervise, or make contributions to the project.

Table of Contents: The purpose of a table of contents is to provide an overview of the material included in the report together with the sequence of presentation. The table of contents includes the following major divisions of the report: the introduction, the chapters with their subsections, and the reference

and appendix. Page numbers for each of these divisions should be given. Care should be exercised that titles of chapters and captions of subdivisions within chapters correspond exactly with those included in the body of the report. Most of the word processing packages are capable of generating table of contents automatically by properly formatting the document.

Abstract: An abstract consists of the following parts:
A short statement of the problem/need, a brief description of the methods and procedures used for research/design, and a condensed summary of the findings/outcome of the project. Normally, the length of abstract should range from 200 to 400 words.

Nomenclature: Nomenclature is not always necessary, but for report in which many mathematical symbols are used, it is preferable to include such a section in the report.

Chapter 1: Introduction: An introduction should be written with considerable care with two major aims in mind. Firstly, describe the problem in a suitable context. Secondly, stimulate readers' interest. An introductory chapter may follow the following structure:

1. Background: You may include a description of an engineering problem and literature review.
2. Issues to be addressed or problems to be resolved in your project.
3. Objectives of the project.
4. Major work outline.

Chapter 2 – P: The main content of the report:
The main content may contain chapters such as:

- Theoretical consideration
- Design details, if applicable
- Experimental details (including error analysis), if applicable
- Simulations / design analysis, if applicable
- Results and discussions

In writing the main body of the report, there are certain general principles, which should be followed:

- Organise the presentation of the argument or findings in a logical and orderly way in order to achieve the objectives stated in Introduction.
- Substantiate arguments or findings based on your data or those available in the literature.
- Be accurate in documentation.

In writing report, every effort should be made to write clearly and forcefully within a logical framework. This framework is provided in the report by a division of the material into appropriate chapters.

Figures, graphs, images, tables and appendices should be identified in the text in a proper numerical sequence, e.g. Figure 1, Table 1, etc.

Chapter Q: Conclusions and possible recommendations for future work: This section serves the important function to conclude the whole report. In summary form, the developments of the previous chapters should be succinctly restated, important findings/outputs discussed and conclusions drawn from the whole project. In addition, you may list unanswered questions/incomplete work that have occurred in the course of the project and which require further research/additional work beyond the limits of the project being reported. The conclusion should leave the reader with the impression of completeness and of positive gain. In addition, provide possible recommendations for future work or

further development.

References: This is a list of the references referred to in the report (do not include references which are not referred to the text of the report). The APA referencing system should be adopted and used consistently throughout report. You may refer to the referencing style section for more details.

Appendices: It is usual to include in an appendix such materials as original data, tables that present supporting evidence, tests that have been constructed by the research, additional information related to design work carried out, parts of documents or any supportive evidence that would detract from the major line of argument and would make the body of the text unduly large and poorly structured. Each appendix should be clearly separated from the next and listed in the table of contents.

Log sheets: Log sheets prepared for each major meeting with the supervisor. All the log sheets should be dated. The standard form of log sheets is available in the **Appendix C.**

9.2 Writing Style

The project reports should be written in an academic writing style. First person references (I, we, us) should NOT be used unless it is appropriate.

9.3 Layout and Format requirements

- Typed, A4 paper size
- Font: Times New Roman font size 12
- Spacing: Single line spacing;
- Page numbers should be indicated at the bottom of each page;
- All sections and subsections should be numbered and given a title;
- Legibility of all submitted copies (drafts and final submission) should be ensured, i.e., text and images of the copy should be ready for reproduction from a black and white printer;

9.4 Referencing Style

The APA referencing system should be adopted and used consistently throughout all written reports. References must be listed in alphabetical order by author. Some examples are available from the following website: https://www.polyu.edu.hk/elc/independent-learning/language-resources/apa/.

<u>*Example: Journal article, one author*</u>
Ku, G. (2008). Learning to de-escalate: The effects of regret in escalation of commitment. *Organizational Behaviour and Human Decision Processes, 105(2*), 221-232.

- "Ku" is the family name of the author, and "G" is the initial of the author's first name (e.g. George is the first name; therefore, "G" is the initial).
- "Learning to de-escalate: The effects of regret in escalation of commitment" is the article title.
- "Organizational Behaviour and Human Decision Processes" is the journal title.
- "105(2)" are the volume and issue number.
- "221-232" are the page numbers.

EXAMPLES OF REFERENCES BY TYPE		
	In a reference list	In-text citation
1	**Journal article (print version), one author** Ku, G. (2008). Learning to de-escalate: The effects of regret in escalation of commitment. *Organizational Behaviour and Human Decision Processes*, *105(2),* 221-232.	(Ku, 2008)
2	**Journal article (print version), two authors** Lefever, M., & Withiam, C. (1998). Curriculum review: How industry views hospitality education. *Cornell Hotel and Restaurant Administration Quarterly*, *39(4),* 70-78.	(Lefever & Withiam, 1998)
3	**Journal article (print version), more than two authors** Weaver, P., Choi, J., & Kaufman, T. (1997). Question wording and response bias: Students' perceptions of ethical issues in the hospitality and tourism industry. *Journal of Hospitality and Tourism Education*, *9(2),* 21-26.	(Weaver, Choi & Kaufman, 1997) Then subsequently (Weaver, et al., 1997)
4	**Article from an Internet-only journal (i.e. no print version)** Hirtle, P. B. (2008, July-August). Copyright renewal, copyright restoration, and the difficulty of determining copyright status. *D-Lib Magazine*, *14*(7/8). Retrieved from http://www.dlib.org/dlib/july08/hirtle/07hirtle.html	(Hirtle, 2008)
5	**Conference paper online** Bochner, S. (1996, November). *Mentoring in higher education: Issues to be addressed in developing a mentoring program.* Paper presented at the Australian Association for Research in Education Conference, Singapore. Retrieved from http://www.aare.edu.au/96pap/bochs96018.txt	(Bochner, 1996)
6	**Newspaper article** Hartevelt, J. (2007, December 20). Boy racers. *The Press Newspaper*, P.3	(Hartevelt, 2007)
7	**Newspaper article (no author, print version)** Brexit beyond. (2018, May 5). *The Wall Street Journal*, p. A9. *Note: article title comes first.*	("Brexit beyond," 2018)
8	**Magazine article (print version)** Kluger, J. (2008, January 28). Why we love. *Time*, *171*(4), 54-60.	(Kluger, 2008)
9	**Book with one author** Kidder, T. (1981). *The soul of a new machine*. Boston, MA: Little, Brown & Company.	(Kidder, 1981)
10	**Book with two authors** Frank, R. H., & Bernanke, B. (2007). *Principles of macro-economics* (3rd ed.). Boston, MA: McGraw-Hill/Irwin.	(Frank & Bernanke, 2007)
11	**Book with more than two authors** Krause, K.L., Bochner, S., & Duchesne, S. (2006). *Educational psychology for learning and teaching* (2nd ed.). South Melbourne, Vic., Australia: Thomson.	(Krause, Bochner, & Duchesne, 2006) Then subsequently, (Krause et al., 2006)

EXAMPLES OF REFERENCES BY TYPE		
	In a reference list	In-text citation
12	**Book or report by a corporate author, e.g. organization association, government department** University of Waikato. (1967). *First hall of residence* (Information series No.3). Hamilton, New Zealand: Author. *Note: when the publisher and author are the same, put down "Author" in the publisher field.*	(University of Waikato, 1967)
13	**Book chapter in edited book** Hammond, K. R., & Adelman, L. (1986). Science, values, and human judgment. In H. R. Arkes & K. R. Hammond (Eds.), *Judgment and decision making: An interdisciplinary reader* (pp. 127-143). Cambridge, England: Cambridge University Press.	(Hammond & Adelman, 1986)
14	**Webpages** Statistics New Zealand. (2007). *New Zealand in profile 2007*. Retrieved from http://www.stats/govt.nz	(Statistics New Zealand, 2007)

For more information on APA citation style, please visit

- https://www.waikato.ac.nz/__data/assets/pdf_file/0003/510195/apa-quick-guide.pdf
- http://elc.polyu.edu.hk/referencing/APA.aspx#undefined
- http://www2.elc.polyu.edu.hk/CILL/referenceMachineNet.htm#Example
- https://www.library.cornell.edu/research/citation/apa

9.5 Length of Report

The report should be written such that it can be read and absorbed by an engineer having a basic knowledge of the subject. It should not contain irrelevant material. The length of the main body of the final report should be at least 45 pages; it is advisable not to write a report exceeding 200 pages.

9.6 Assessment of Individual Contribution

Contribution of each student to a project has to be assessed. Each student is required to write their contributions towards the project monthly in the online learning platform. At the end of the project, induvial member is also required to submit a summary of his contribution to the project. A sample is shown in **Appendix D.** The summary should be submitted together with the report. In addition, each student has to conduct a peer assessment. The assessment form is given in **Appendix E**. The peer assessment will be kept confidential.

10 Report Submission Policies

10.1 Submission Timelines and Requirements

The proposal and reports must be submitted online as follows:

Report	Submission Deadline
Formal project proposal	Week 4 – Semester 1
Interim progress report	Week 19 – Semester 1
Final Report	Week 13 – Semester 2

* Please refer to the announcement for the exact submission deadline.

Note: Students should submit the proposal, entire progress report and final report to the Turnitin system, including title page, declaration form, supervision log sheet etc. Turnitin submission link will be provided in online learning platform. Late submissions will not be accepted.

10.2 Late Submission

Late submission of proposal and reports (interim report and/or final report) will NOT be accepted.

Interim Report

In principle, late submission equals 0 mark.

If a student group is unable to submit on time because of an unforeseeable reason, the group may apply for a late submission with documentation endorsed by the project supervisor. Permission is subject to the approval of the Project Coordinator/ Programme Leader. Should late submission be granted, the group concerned would receive the actual grade they attained on the report submitted.

Final Report

In principle, late submission equals 0 mark.

If a student group is unable to submit on time because of an unforeseeable reason, the group may apply for a late submission with documentation endorsed by the project supervisor. Granting of late submission is subject to the decision of Project Coordinator / Programme Leader in consultation with Project Supervisor. Should late submission be granted, the group concerned would receive the actual grade they attained on the Final Report submitted.

11 Re-assessment / Retake

Student groups that are considered marginally failed may be given extra time to complete the project work to the satisfaction of Project Supervisor subject to the decision of SARP. If such an extension is given to a student group, all the students in the group concerned would be given a result of "I" (i.e. Incomplete), which is interpreted as "Assessment to be completed".

SARP would decide a deadline for completion of the reassessment. The student would not be required to pay additional fees.

The highest grade to be given for re-assessed projects should be grade "C". The results of re-assessed projects would be finalized in the next SARP meeting.

Student groups that have not met the subject requirement and were not given a re-assessment opportunity would fail the subject. They would be required to retake the subject and pay the corresponding tuition fees.

Student groups that have failed the Final Year Capstone Project may be requested by SARP to attempt a new project topic.

12 Plagiarism Policies and Turnitin Report

PolyU SPEED has detailed policy on student's academic integrity, which include plagiarism and the use of generative artificial intelligence. The following are the related issues stipulated clearly in the Student Handbook:

12.1 Violations of Academic Integrity

The School views any violations of academic integrity a serious disciplinary offence because it seriously undermines the value of the teaching, learning and research of the School. For students who cheat in their academic work, they limit their learning because they have not undergone the learning experience intentionally designed to help them to gain the specific knowledge or skills. At the same time, if some students cheat in the assessment and get a good grade, it is not fair to the others who are honest and do their own work. Finally, academic dishonesty will tarnish the reputation of the programme and its graduates and in the long run, diminish the value of the degree or education from the School.

Students should therefore refrain from committing any acts of academic dishonesty. One of such acts which are highly related to research projects is Plagiarism.

12.2 About Plagiarism

Plagiarism refers to the act of using the creative works of others (e.g. ideas, words, images or sound, etc) in one's own work without proper acknowledge of the source. Handing in the same assignment for more than one subject is considered as self-plagiarism. According to the Webster's Ninth New Collegiate Dictionary (1987), to 'plagiarise' means

> [To] steal and pass off (the ideas or words of another) as one's own: [to] use (a created production) without crediting the source: [to] commit literary theft: [to] present as new and original an idea or product derived from an existing source.

The University views plagiarism, whether committed intentionally or because of ignorance or negligence, as a serious disciplinary offence. Excuses such as "not knowing that this is required" or

"not knowing how to do it" will not be accepted. It is the student's responsibility to understand what plagiarism is, and take action steps to avoid plagiarism in their academic work. The golden rule is: "if in doubt, acknowledge".

Avoiding Plagiarism
Students are required to submit their original work and avoid any possible suggestion of plagiarism in the work they submit for grading or credit. Below are some suggestions on how you can avoid plagiarism in your own work:

Use Sources with Care and Respect
- Take careful notes so that you know where you got your information;
- Keep track of all the sources you have used for each assignment;
- Cite all your sources in your finished work, distinguishing carefully between your own ideas/ work and those taken from others;
- Include all your sources in your Reference or Bibliography section, normally included at the end of the paper.

Find out the Expectations of Your Teacher
- Different disciplines or professions may have slightly different conventions for citation and referencing. Ask your teacher for the specific citing and reference system or conventions used in your chosen profession/ discipline;
- Ask your teacher what types of collaborations and help is permitted for the specific assignment.

Develop Your Academic Skills
- Plan your academic work carefully and start early so that you have time to do your own work;
- Make a work schedule for your work and try to keep to it;
- Study resource materials and, if appropriate, attend courses or workshops to continually improve your skills in referencing and academic writing.

Be Honest, and Always Do Your Own Work
The School may take disciplinary actions against students when there is evidence of collusion between individuals. The work of others, which is included in the assignment must be attributed to its source (a full bibliography and a list of reference must be submitted). Failure to observe such requirements will definitely lead to serious consequence for your study on this subject and registration at School.

12.3 Turnitin Report

Turnitin database will be used to check for plagiarism. All students should note that allowing another student to copy their work is a serious breach of academic discipline. If the submissions include a high percentage of work that is copied from other sources, a fail grade will be awarded.

Turnitin User Guide for Students
The Turnitin User Guide for students is available at http://portal.cpce-polyu.edu.hk. Please login and then choose "VeriGuide and Turnitin User Guides" for details.

Trial Submission
Before the final submission, students could submit the progress report and final report via "Trial Submission" to check the similarity index more than once. The 2nd trial report will not be checked

against the 1st trial report submitted by the same student; therefore, the similarity index will not be affected by multiple submissions via "Trial Submission".

After reading the report, you might need to revise your work accordingly. The purpose of revising your work is to make it better and in particular, plagiarism-free.

The following are some tips on how you can review and revise your work effectively.

Tip 1 When reviewing the matching texts and their sources, check carefully if proper acknowledgements are given to the sources you used.

- ✓ If yes, then you are fine. Remember to include all the sources you used in the reference or bibliography list as well.
- ✘ If no, then cite the sources properly using the specified referencing style.

Tip 2 You can use others' words verbatim as direct quotations, i.e., embedded quotes with quotation marks or block quotes with indentation; and you still need to cite the sources.

- **!!!** Bear in mind that excessive use of quotations is not encouraged as it is not a good writing practice. Using too many quotations makes your work look weak as you are simply patching together others' words without much input of your own.

Tip 3 You can paraphrase or summarize others' words or ideas in your own words, and cite them properly.
You should build your ideas or thoughts upon others' work and express them in *your own words.*

The Similarity Index shows the percentage of matching or highly similar text found in a submitted paper compared against the sources stored in the system's database. It ranges from 0% to 100%.

According to the "Instructor Guide to Using Turnitin Assignment with LEARN@PolyU" developed by the Working Group on Promoting Students' Academic Integrity at PolyU, there is no rigid percentage to determine whether plagiarism exists.

Supervisors will consider both the "Similarity Index" and "Match Overview" to see the details of the matches and make professional judgments when detecting plagiarism.

12.4 Consequences of Plagiarism

- Students who are found guilty of offences related to plagiarism will be subject to the penalty of having their subject result disqualified and being given a failure grade with a remark denoting that it is due to academic dishonesty. The remark will be appropriately shown on the students' record and on documents such as assessment result notification and transcript of studies until they leave the School. These students will also be subject to the penalty of the lowering of their award classification by one level upon graduation. For example, for students who should be awarded a Third class Honours degree, they will be downgraded to a Pass-without-Honours. The minimum of downgraded classification will be kept at a pass.

- Students who are found guilty of the alleged offences will be put on "disciplinary probation". The status of "disciplinary probation" will be shown on the students' record and on documents such as assessment result notification, transcript of studies and testimonial during the probation period,

which will normally be one year unless otherwise decided by the CPCE Student Discipline Committee. Students who have been put on disciplinary probation will be deprived of certain privileges. They shall not receive honour from PolyU/ CPCE/ SPEED (such as scholarships/ awards/ prizes) or engage in activities which represent PolyU/ CPCE/ SPEED (such as outstanding students/ student ambassadors/ taking up leadership roles within the University/ School).

- Students may also be given lower priority in funding and subsidies for student projects, courses/ activities, overseas academic exchange, internship jobs, mentorship programmes, overseas WIE, etc.

Appendix A - Sample Title Page

School of Professional Education and Executive Development

The Hong Kong Polytechnic University

SEHS4654 Capstone Project

Bachelor of Engineering (Honours) in Electrical Engineering

CAPSTONE PROJECT TITLE

Formal Project Proposal / Interim Progress Report / Final Report
(Delete as appropriate)

Group No	
Student No.	**Full Name**

Supervisor/s: XXX

Submission date: dd/mm/yyyy

Appendix B –Capstone Project Declaration Form

CAPSTONE PROJECT REPORT DECLARATION FORM

We hereby declare that the Capstone Project report, the title of which is _____, submitted on (date) for PolyU SPEED programme (Bachelor of Engineering (Honours) in Electrical Engineering), is our work and that, to the best of our knowledge and belief, it reproduces no material previously published or written, nor the work of other students, nor material that has been accepted for the award of any other degree or diploma, except where due acknowledgement has been made in the text.

Signed by:

1. Full name (Student No.) (Signature)
2. Xxx
3. xxx

Date:_____

Note: This declaration form must be handed-in together with the ***Final Report****.*

Appendix C - Sample Supervision Log Sheet

SEHS4654 Capstone Project
Log of the _____ Meeting *(eg. 1st Meeting, to be **filled in by student**)*
(Student should submit the log sheets online monthly, and with the final report.)

Date and Time of Meeting:	
Venue and Duration:	
Brief description of work done:	
Agenda (Any issues, concerns, problems for discussion):	
Agreed tasks or action plan for next meeting:	
Students' Names:	Signatures:

Appendix D - Sample Summary of Contribution

SEHS4654 Capstone Project

Summary of Contribution

Project: Study of emissions from a diesel engine

Name of Student: Student 1 (writer)

Names of group-mate(s): Groupmate 1 and Groupmate 2

This project involves the following tasks: (1) searching for background material; (b) setting up a test rig for conducting the tests; (c) measurement of diesel particles with a Scanning Mobility Particle Sizer based on the R49 13-mode test method; (d) measure the exhaust gas concentrations with different gas analyzers based on the R49 13-mode test method; (e) assess the effectiveness of a diesel oxidation catalyst in reducing pollutant emissions from the engine, again based on the R49 13-mode method; (f) evaluation of the particulate data; (g) evaluation of the exhaust gas data based on SAE standard; (h) writing the report.

We worked together in searching for background material and I concentrated on reviewing emission testing methods as contained in section 1.5 of Chapter 1. Section 1.5 was written by me.

I was mainly responsible for designing and setting up the test rig as described in Chapter 2 of the report; and Chapter X was written by me. The design was carried out in accordance SAE Standard XX.

The experiments were carried out by my group-mates. I was responsible for analysing the experimental data based on the R49 13-mode test procedure. Chapter 3 on Methodology of data analysis was written by me. Also, results and discussion contained in Chapter 4 and Chapter 5 were done by me and the corresponding sections in these chapters were written by me (sections 4.3 and 4.4; sections 5.3 and 5.4).

We worked together on the Conclusions and we have equal contribution in this part.

Appendix E – Confidential Peer Assessment Form

SEHS4654 Capstone Project
Peer Assessment Form
CONFIDENTIAL

You are required to assess your contribution and your peers' contribution in each task of the project. The contribution rating varies from 0% to 100%. 0% means no contribution at all, while 100% means all the work are carried out by the student concerned.

The information provided will be treated with strict confidentiality and will not be revealed to any of your group-mates or classmates.

Group No.	
Project Title	

Project Tasks	**Distribution of contribution (total 100% for each item)**			
	Name of Assessor (your name):	**Name of Groupmate 1:**	**Name of Groupmate 2:**	**Total**
Formal project proposal writing				100%
Interim progress report writing				100%
Final report writing				100%
Presentation and demonstration				100%
Attitude and overall contribution for the project				100%

Any other comments:

Name of Assessor: _____

Signature of Assessor: _____

Date: _____